# Revision-Note Review Worksheet for Expert Rater

This booklet contains one student revision note to review. It is a take-home revision note prepared by a teacher for a JSS1 Mathematics class on Three dimensional figures: properties of cylinders and spheres, and volume. Judge how well it covers the intended objectives and curriculum, follows the teacher's lesson plan, stays on topic and is conceptually correct, and is clear and useful for the students.

## Lesson and revision-note context

| | |
|---|---|
| **Subject** | Mathematics |
| **Grade** | JSS1 |
| **Term** | Third Term |
| **Week** | 5 |
| **Topic** | Three dimensional shapes II |
| **Subtopic** | Three dimensional figures: properties of cylinders and spheres, and volume |
| **Theme** | — |
| **Curriculum objectives** | • identify properties of cylinders and spheres<br>• find the volume of a cube, a cuboid and a prism |
| **Previous knowledge** | • properties of cubes and cuboid<br>• properties of pyramids and cones |
| **Lesson sequence** | • identify properties of cylinders and spheres — the teacher displays shapes of cylinders and spheres; the student identify the properties; the teacher draws the shapes on the board and adds to the properties already mentioned by the students. Set for the students: 1. How many circular faces does a solid cylinder have? 2. A cylinder has how many faces.<br>• find the volume of a cube, a cuboid and a prism — the teacher explains how to derive the formula for volume of cube and cuboid, and works the volume of a prism as area of base times height; the student listen and ask questions where they do not understand; the student copy worked examples into their notes |
| **Instructional materials** | • shapes of cylinders and spheres. |
| **Evaluation** | • Calculate the volume of a cuboid 8cm long, 4cm wide and 3cm high.<br>• Calculate the volume of cuboid with dimensions 5m by 2m by 3m.<br>• What is the volume of a cube of 4cm × 4cm × 4cm.<br>• A rectangular tank hold 6m long by 2m wide holds $36m^3$ of water. How deep is the water in the tank. |

## Lesson objectives

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|---|---|
| **1.** | identify properties of cylinders and spheres |
| **2.** | find the volume of a cube, a cuboid and a prism |

Read the student revision note on the following pages, then complete the rating sheet at the end of this booklet.

# STUDENT REVISION NOTE

**Topic:** Three dimensional figures: properties of cylinders and spheres, and volume

**Class:** JSS1 **Subject:** Mathematics **Week:** 5

## Section 1: Cylinders and Spheres: Their Properties

Two of the most familiar three-dimensional figures around you are the cylinder and the sphere. A can of drink is a cylinder; a football or a ball of wool is a sphere.

Start with the cylinder. Consider the cylinder in fig-01. A solid cylinder has two circular faces: one on top and one at the bottom. Joining them together around the outside is a single curved face that rolls around the whole body. So altogether a solid cylinder has:

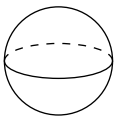
- two circular faces and one curved face, making **three faces** altogether;
- **no vertex** — there is no sharp corner or point anywhere on it;
- **two curved edges**, the two circles where each circular face meets the curved face.

Now consider the sphere in fig-02. A sphere is a perfectly round figure, like a ball. Its properties are simpler still:

- it is **symmetrical from all directions** — however you turn it, it looks the same;
- it has **no edge and no vertex** — there are no straight lines, corners or points on it;
- it has **one curved surface** that covers the whole figure.

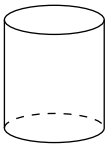
The easiest way to tell a cylinder from a sphere is to count the faces and edges: a cylinder has three faces and two curved edges, while a sphere has just one curved surface and nothing to count as an edge or vertex.

**Visual**



A sphere drawn as a circle with an ellipse across its middle to show the curved surface.

**Visual**



A cylinder drawn upright: an ellipse for the top face, two vertical sides falling from its ends, and a curve closing the base.

**Practice**

1. How many circular faces does a solid cylinder have?
2. A cylinder has how many faces altogether?
3. Does a sphere have any edges or vertices? Explain.

## Section 2: Volume of a cube, a cuboid and a prism

Volume is the amount of space a shape takes up. For a cuboid — a box with a rectangular base — you find the volume by multiplying the length, the breadth and the height:

$$\text{Volume of a cuboid} = \text{length} \times \text{breadth} \times \text{height} = lbh$$

You can also think of it as the area of the base times the height. Since the base is a rectangle of area $l \times b$, multiplying by the height $h$ gives $lbh$.

A cube is a cuboid whose length, breadth and height are all the same. If each side is $l$, then:

$$\text{Volume of a cube} = l \times l \times l = l^3$$

For a prism — a solid with a constant cross-section — the volume is the area of the base (the end face) times the height:

$$\text{Volume of a prism} = \text{area of base} \times \text{height}$$

You use whichever end face is constant, find its area, then multiply by how far the prism runs back.

**Example 1. Find the volume of a cuboid of length 3 cm, breadth 2 cm and height 8 cm.**

$$\text{Volume} = L \times b \times h$$

Length $= 3$ cm, breadth $= 2$ cm, height $= 8$ cm.

$$\text{Volume} = 3 \text{ cm} \times 2 \text{ cm} \times 8 \text{ cm} = 48 \text{ cm}^3$$

So the cuboid holds 48 $\text{cm}^3$.

**Example 2. The volume of a cuboid is 24 $\text{cm}^3$. If its base is 6 cm by 2 cm, find its height.**

$$\text{Volume} = L \times b \times h$$

Length $= 6$ cm, breadth $= 2$ cm, height $= h$ (unknown).

$$24 \text{ cm}^3 = 6 \text{ cm} \times 2 \text{ cm} \times h$$
$$24 \text{ cm}^3 = 12 \text{ cm}^2 \times h$$

Divide both sides by 12 $\text{cm}^2$:

$$\frac{24 \text{ cm}^3}{12 \text{ cm}^2} = \frac{12 \text{ cm}^2 \times h}{12 \text{ cm}^2}$$
$$2 \text{ cm} = h$$

So the height is 2 cm.

**Example 3. Find the volume of a cube of side 3 cm.**

$$\text{Volume of a cube} = l^3$$
$$V = (3 \text{ cm})^3 = 3 \text{ cm} \times 3 \text{ cm} \times 3 \text{ cm} = 27 \text{ cm}^3$$

**Example 4. Find the volume of the triangular prism**, where the triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

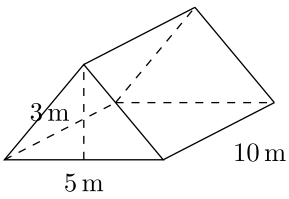
First find the area of the triangular base:

$$\text{Area of triangle} = \tfrac{1}{2} \times \text{base} \times \text{height} = \tfrac{1}{2} \times 5 \times 3 = 7.5 \text{ m}^2$$

Then multiply by the length of the prism (10 m):

$$\text{Volume of prism} = \text{area of base} \times \text{height} = 7.5 \text{ m}^2 \times 10 \text{ m} = 75 \text{ m}^3$$

**Visual**




A triangular prism lying on its rectangular face. The triangular end has a base of 5 m and a height of 3 m, and the prism runs back 10 m.

**Practice**

1. A cuboid is 8 cm long, 4 cm wide and 3 cm high. Find its volume.
2. A cuboid has dimensions 5 m by 2 m by 3 m. Calculate its volume.
3. A cube measures 4 cm by 4 cm by 4 cm. What is its volume?
4. A rectangular tank is 6 m long and 2 m wide, and it holds 36 $\text{m}^3$ of water. How deep is the water in the tank?

## Check your work

Attempt every question before you read this page.

## Cylinders and Spheres: Their Properties

1. A solid cylinder has two circular faces (one on top and one at the bottom).

   Look at a solid cylinder: the top is one circle and the bottom is another circle, so there are two circular faces in total.
2. A cylinder has three faces altogether: two circular faces and one curved face.

   Count them: one circular face on top, one circular face at the bottom, and the curved face wrapping around the outside. That gives $1 + 1 + 1 = 3$ faces.
3. No. A sphere has no edge and no vertex — it is one continuous curved surface with no corners or straight edges.

   A sphere is perfectly round, like a ball. Because it is smooth all over, there is no straight line (edge) and no corner or point (vertex) anywhere on it.

## Volume of a cube, a cuboid and a prism

1. $8 \times 4 \times 3 = 96\ \text{cm}^3$

   Volume of a cuboid $= L \times b \times h = 8\ \text{cm} \times 4\ \text{cm} \times 3\ \text{cm} = 96\ \text{cm}^3$.
2. $5 \times 2 \times 3 = 30\ \text{m}^3$

   Volume of a cuboid $= L \times b \times h = 5\ \text{m} \times 2\ \text{m} \times 3\ \text{m} = 30\ \text{m}^3$.
3. $4^3 = 64\ \text{cm}^3$

   Volume of a cube $= l^3 = (4\ \text{cm})^3 = 4 \times 4 \times 4 = 64\ \text{cm}^3$.
4. 3 m

   Volume $= L \times b \times h$, so $36\ \text{m}^3 = 6\ \text{m} \times 2\ \text{m} \times h = 12\ \text{m}^2 \times h$. Divide both sides by $12\ \text{m}^2$: $h = \frac{36\ \text{m}^3}{12\ \text{m}^2} = 3\ \text{m}$.

## Rating sheet

Mark one score per criterion, then use the notes section for strengths, concerns, and recommendations. The lesson objectives are on the first page for reference.

| | | | |
|---|---|---|---|
| **Revision note ID** | cand-2d465c56 | **Date** | |
| **Subtopic** | Three dimensional figures: properties of cylinders and spheres, and volume | **Rater** | |

The reference sheet includes the teacher's evaluation questions. A pack that reuses them, reworded or verbatim, may be credited for matching how the lesson was assessed — but reuse on its own does not establish taught-lesson alignment. Rate alignment on the whole of it: the objectives covered, the order taught, what the activities had the students do, the method, and the expected depth.

| **Criterion** (select one score per row) | **1** Poor | **2** Weak | **3** Adequate | **4** Good | **5** Excellent |
|---|---|---|---|---|---|
| **Objective and curriculum alignment** Does the revision note cover the intended lesson objectives and curriculum target? | ○ | ○ | ○ | ○ | ○ |
| **Taught-lesson alignment** Does it reflect the taught lesson — its sequence, emphasis, activities, and expected depth? | ○ | ○ | ○ | ○ | ○ |
| **Scope and conceptual correctness** Does it stay within the expected lesson boundary and avoid conceptual/mathematical errors? | ○ | ○ | ○ | ○ | ○ |
| **Student-facing clarity and usefulness** Can JSS1 learners understand and use it meaningfully as take-home review material? | ○ | ○ | ○ | ○ | ○ |

| Strengths | Concerns | Recommendation |
|---|---|---|
| | | |

## Flag any problems (optional)

Tick all that apply. These are separate from the scores above — leave blank if none apply.

| | |
|---|---|
| ☐ | **Misses a required objective** — Skips something the lesson was meant to teach. |
| ☐ | **Goes beyond this lesson** — Includes material outside this lesson's scope (e.g. next week's topic). |
| ☐ | **Wrong or misleading content** — Contains a mathematical or factual mistake. |
| ☐ | **Too hard for JSS1 to follow** — Likely confusing for the students. |
| ☐ | **Other:** _____ |